

Machine Learning Models

1. Airbnb Price Prediction Model

A machine learning regression approach was developed to predict Airbnb listing prices based on property characteristics and host-related features. Before model training, the dataset underwent several preprocessing steps, including feature selection, data transformation, scaling, and train-test splitting.

The input variables used for modelling consisted of selected numerical and categorical features related to Airbnb listings, such as property attributes, location information, availability details, host characteristics, and listing performance indicators. Feature selection was performed using correlation analysis, where numerical features with an absolute correlation value greater than **0.05** with the target variable were retained (Particularly, the features selected are: 'accommodates', 'bedrooms', 'host_years', 'host_identity_verified_t', 'instant_bookable_f', 'amenity_count', 'review_scores_location', 'host_total_listings_count', 'host_acceptance_rate', 'latitude', 'host_response_rate', 'instant_bookable_t', 'host_identity_verified_f'). Redundant variables and variables that could introduce data leakage or did not contribute to prediction, such as listing_id, host_id, price, price_usd, and exchange_rate, were removed before model training.

After feature selection, the dataset was prepared for modelling by separating the independent variables (**X**, selected features) and the target variable (**y**, Airbnb price). Features were standardised using a scaler to ensure that variables with different ranges contributed proportionally during model training.

The processed dataset was divided into training and testing sets using a train-test split strategy. The training set was used to build and optimise the models, while the testing set was reserved for evaluating the model's ability to generalise to unseen Airbnb listings. This approach helps reduce overfitting and provides a more reliable assessment of predictive performance.

Four machine learning regression models were developed and evaluated to predict Airbnb listing prices, including **Random Forest**, **XGBoost**, **Gradient Boosting**, and **Extra Trees**. The models were assessed using two evaluation metrics: **Mean Absolute Error (MAE)** and **R² score**. MAE measures the average difference between the predicted and actual prices, where lower values indicate better prediction accuracy. The R² score represents the proportion of variance in Airbnb prices explained by the model, with higher values indicating better predictive performance.

Model	MAE	R ²
Random forest	0.38	0.75

XGBoost	0.41	0.72
Gradient Boosting	0.40	0.73
Extra Trees	0.39	0.73

Based on the evaluation results, **Random Forest** achieved the best overall performance, obtaining the lowest MAE value of **0.38** and the highest R^2 score of **0.75**. This indicates that the model produced the most accurate predictions and explained approximately 75% of the variation in Airbnb prices. The strong performance of Random Forest can be attributed to its ensemble learning approach, which combines multiple decision trees to capture complex nonlinear relationships between listing characteristics and price.

The **Extra Trees model** achieved a comparable performance with an MAE of **0.39** and an R^2 score of **0.73**. Although it slightly underperformed compared with Random Forest, it still demonstrated strong predictive capability by reducing variance through increased randomness during tree construction.

The **Gradient Boosting model** obtained an MAE of **0.40** and an R^2 score of **0.73**, showing that sequentially improving weak learners can effectively capture patterns in Airbnb pricing data. However, its performance was slightly lower than the Random Forest model, suggesting that the dataset may contain complex interactions that benefit more from bagging-based approaches.

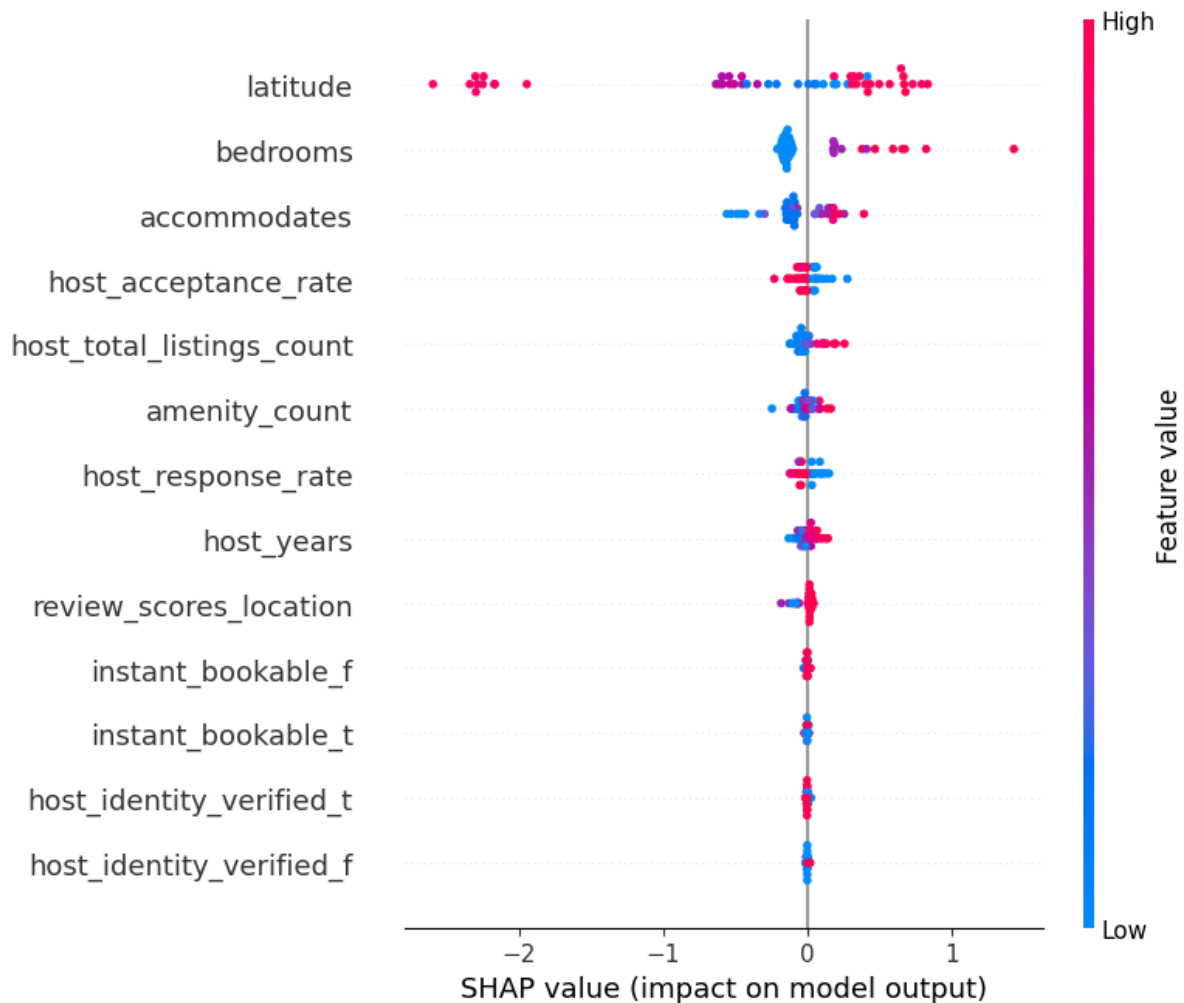
The **XGBoost model** achieved an MAE of **0.41** and an R^2 score of **0.72**, making it the lowest-performing model among the tested approaches. Although XGBoost is widely used for regression problems due to its ability to handle complex feature interactions, the model may require further hyperparameter optimisation to achieve better performance on this dataset.

Overall, the results demonstrate that tree-based ensemble methods are effective for Airbnb price prediction. Among all evaluated models, **Random Forest was selected as the final model** due to its superior accuracy, lower prediction error, and ability to capture nonlinear relationships between property attributes and pricing factors.

2. SHAP Model Interpretation

To improve the interpretability of the machine learning model, SHAP (SHapley Additive exPlanations) was applied to explain how each feature contributes to the predicted Airbnb listing price. Unlike traditional feature importance, SHAP provides both the magnitude and direction of each feature's influence on individual predictions. The SHAP summary plot illustrates how the machine learning model determines Airbnb listing prices by quantifying the contribution of each feature to individual predictions. Features are ranked according to their average impact on model output. Each dot represents a listing, where the horizontal position indicates the SHAP value (the direction and

magnitude of the feature's influence on the predicted price), while the colour represents the feature value, ranging from low values (blue) to high values (red).



Latitude is the most influential feature in the model, showing the widest distribution of SHAP values (approximately -2.5 to +1.5). The wide spread indicates that geographical location has the strongest impact on predicted prices. However, the relationship is not purely linear, as both high and low latitude values appear across positive and negative SHAP ranges. This suggests that latitude acts as a proxy for location-specific characteristics, such as neighbourhood attractiveness, tourism demand, accessibility, and proximity to major attractions. In other words, properties in highly desirable areas can command significantly higher prices, while listings in less competitive locations may experience lower pricing regardless of other characteristics.

Bedrooms is the second most important feature and demonstrates a clear positive relationship with price predictions. Listings with fewer bedrooms (blue points) generally contribute negatively to the model output, reducing predicted prices, while listings with more bedrooms (red points) increase predicted prices. This indicates that larger properties typically achieve higher nightly rates due to greater accommodation capacity and suitability for groups or families.

Similarly, **accommodates** shows a comparable pattern, where higher capacity listings increase predicted prices. The similarity between bedrooms and accommodates suggests that both variables capture related aspects of property size. Although they provide valuable information, some overlap exists because larger properties generally have both more bedrooms and higher guest capacity.

Compared with location and property size, host-related variables have a substantially smaller influence on price predictions. **Host acceptance rate** shows an unexpected inverse relationship, where higher acceptance rates slightly reduce predicted prices. However, the SHAP values remain close to zero, meaning the effect is relatively weak. This pattern may reflect underlying differences between host segments, such as professional hosts managing multiple lower-priced properties or hosts operating in different market segments. Other host features, including **host response rate**, **host years**, and **host total listings count**, provide only minor adjustments to predictions. While higher listing counts and longer hosting experience may slightly increase prices, their contribution is considerably smaller than location and property characteristics.

Features such as **amenity count**, **instant booking availability**, and **host identity verification** have limited influence on predicted prices. Although a higher number of amenities shows a small positive relationship with price, the overall impact remains minor once location and property size are considered.

The low importance of binary features suggests that these characteristics do not independently determine pricing decisions. Instead, their effect may already be captured indirectly through stronger variables such as location, property type, and capacity.

Overall, the SHAP analysis demonstrates that Airbnb pricing is primarily determined by **location and property capacity** rather than host characteristics or individual amenities. For investors, this highlights the importance of selecting properties in high-demand locations and prioritising listings with greater accommodation capacity.